



Technische Hochschule
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End-to-End Learning vs Modular Architecture

Comparative Insights into Autonomous Driving Systems

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Introduction to Autonomous Driving (AD)



- NHTSA report analysis.
- Importance of AD in transportation
 - Safety
 - Efficiency
 - Optimized traffic flow
 - Reduced emissions

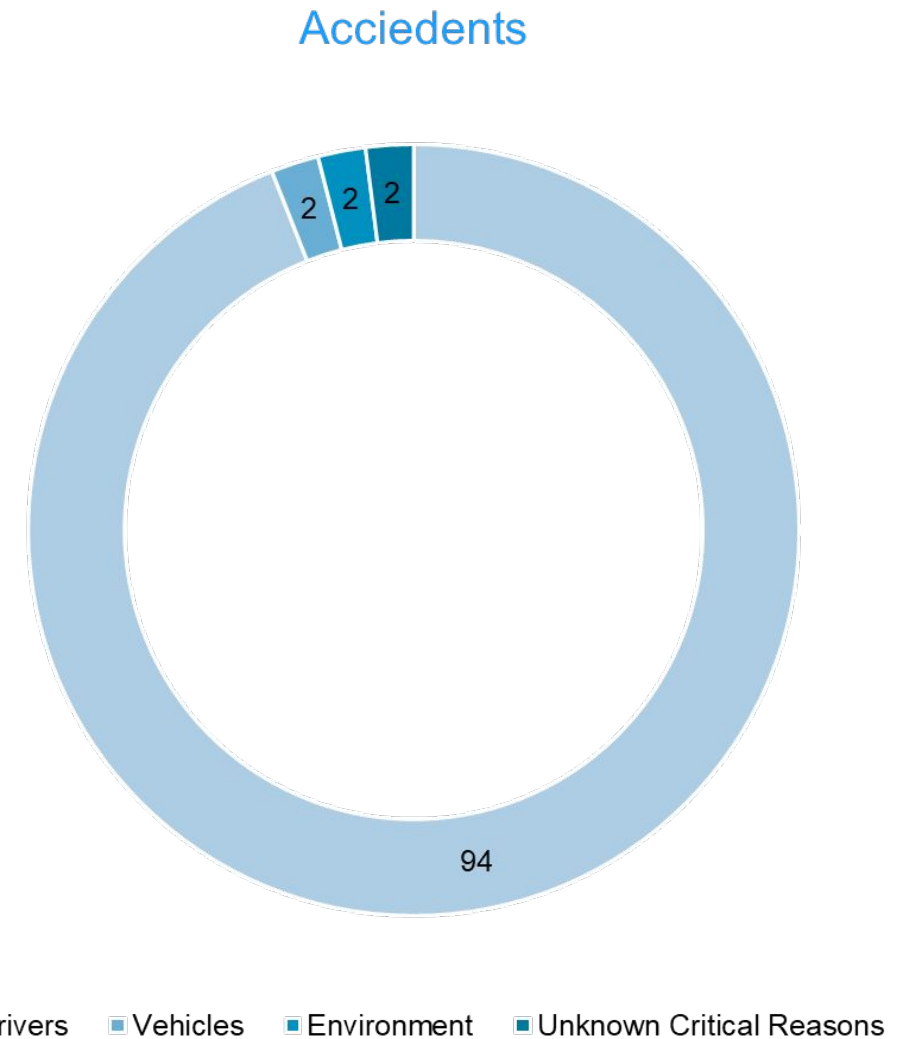
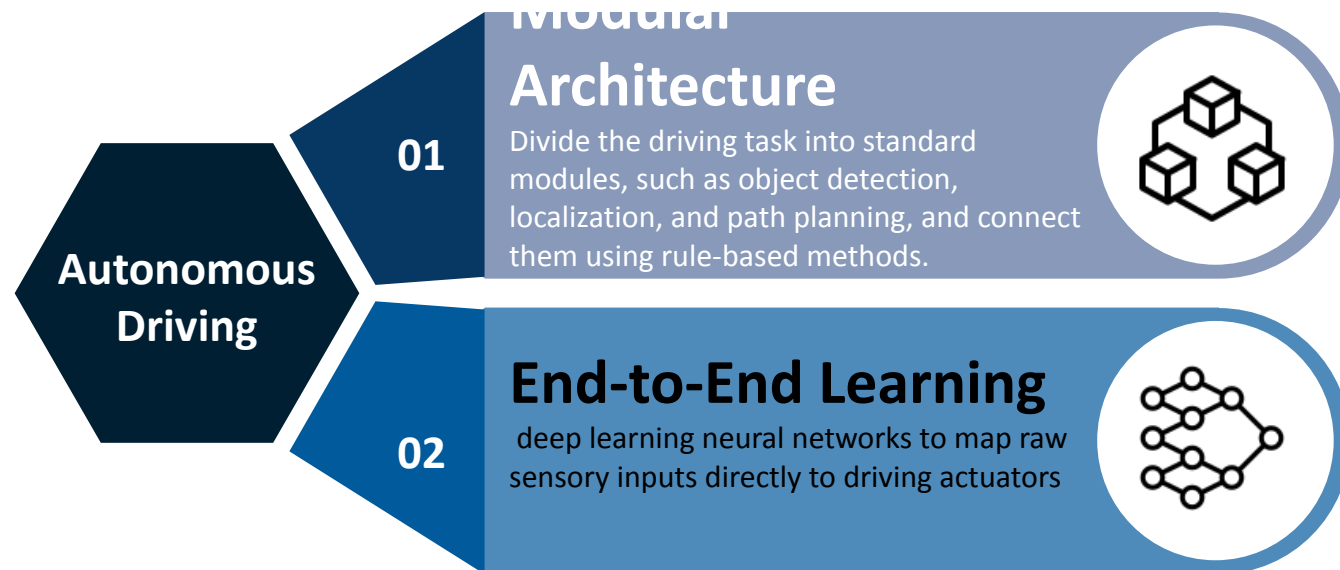


Figure 1. Statistics: 94% of accidents are caused by human error (NHTSA). [1]

Autonomous Driving System Paradigms

1. Modular Architecture

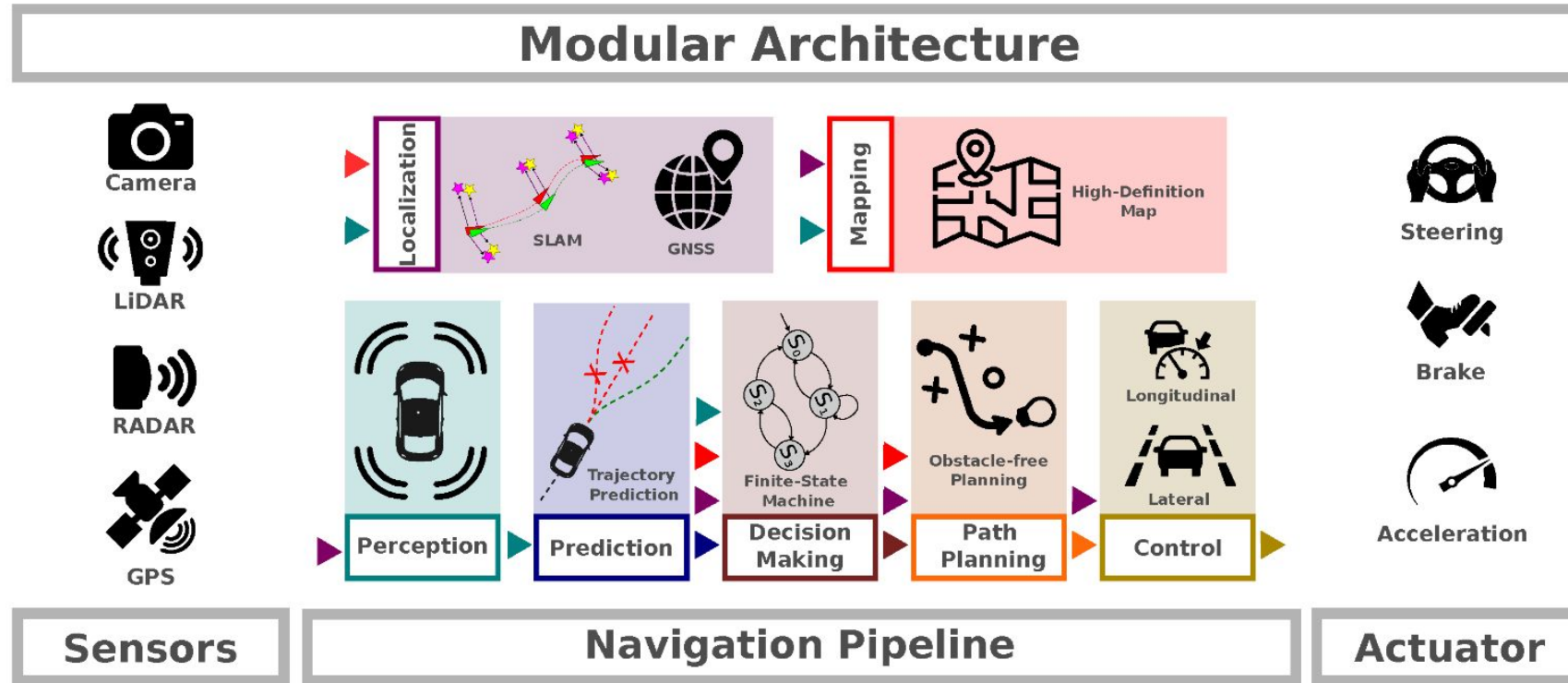


Figure 2. Modular architecture illustration.[2]

- Divides tasks like perception, prediction, decision making, path planning, and control into subsystems.
- Strengths: High interpretability and easier debugging.
- Challenges: Complex integration and error propagation.

Autonomous Driving System Paradigms

2. End-to-End Learning Architecture

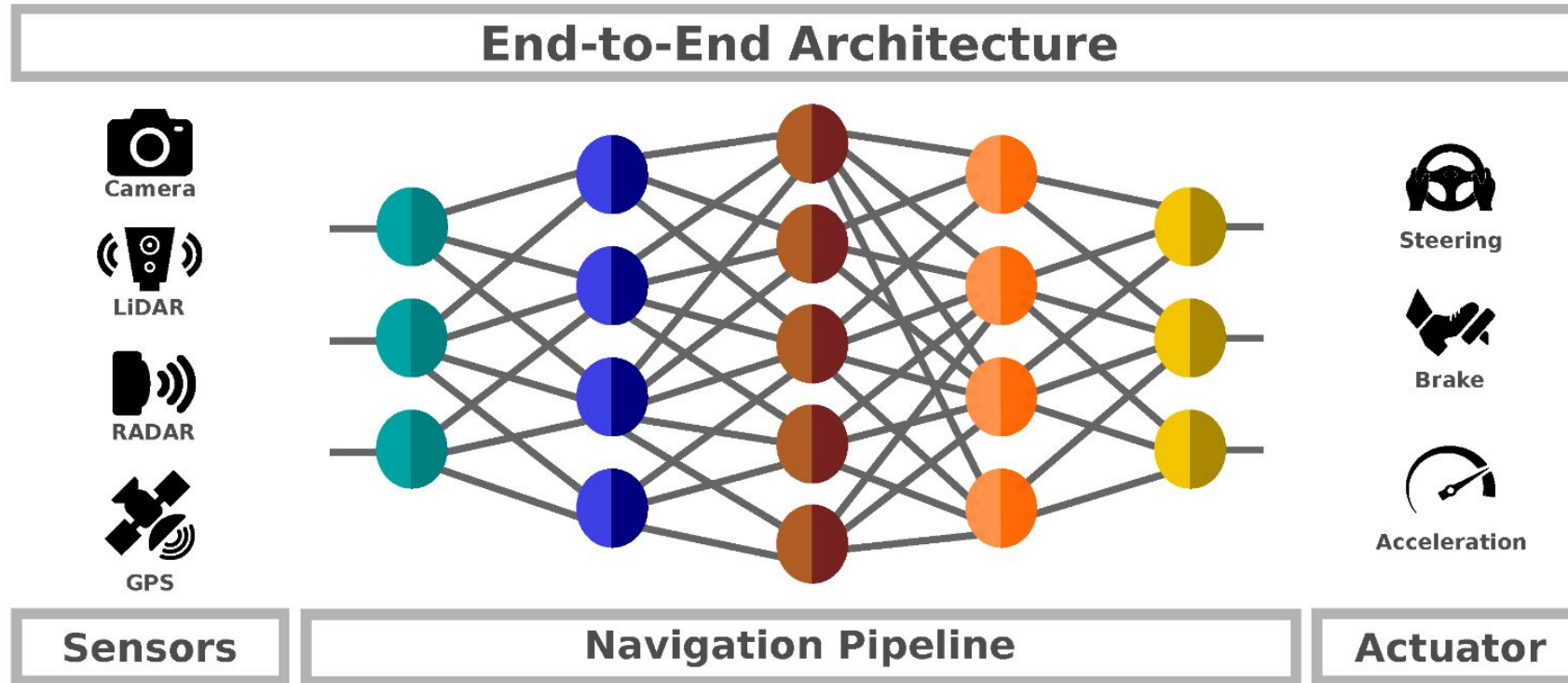


Figure 2. End-to-End Learning architecture illustration.[2]

- Maps sensor data directly to control actions.
- Strengths: Simplicity and adaptability to dynamic scenarios.
- Challenges: Lacks transparency and requires extensive data

Autonomous Driving System Paradigms

2. End-to-End Learning Architecture - Evolution

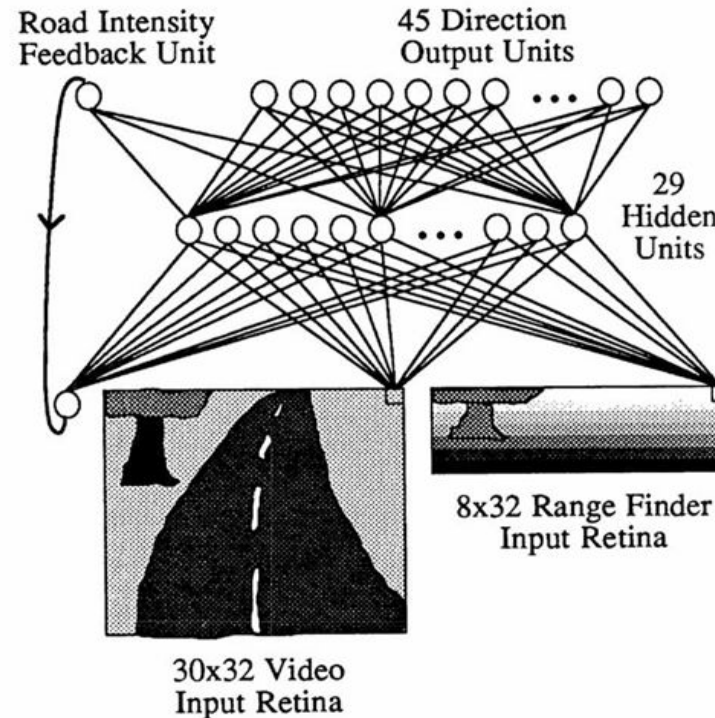


Figure 3. ALVINN architecture illustration.[3]

- *Early systems like ALVINN (1989) demonstrated End-to-End Learning using neural networks for steering control.*
- *Advances in Deep Reinforcement Learning (e.g., DeepMind) have reinvigorated End-to-End Learning.*
- *Modular systems were traditionally favored in industry for their reliability and interpretability.*

Autonomous Driving System Paradigms

3. Comparative Overview



Modular Architecture

VS

End-to-End Learning



Transparency



Flexibility



Error Diagnosis



Data Requirement



Scalability



Autonomous Driving System Paradigms

3. Comparative Overview - Challenges in Both Paradigms

- **Modular Systems:**
 - Integration challenges due to inter-module dependencies.
 - Propagation of errors through the system can lead to failures.
 - Limited adaptability to new or unforeseen conditions.
- **End-to-End Learning:**
 - Lacks interpretability, acting as a "black box."
 - Requires extensive, diverse datasets to handle rare scenarios.
 - Struggles with edge cases and safety-critical decisions.

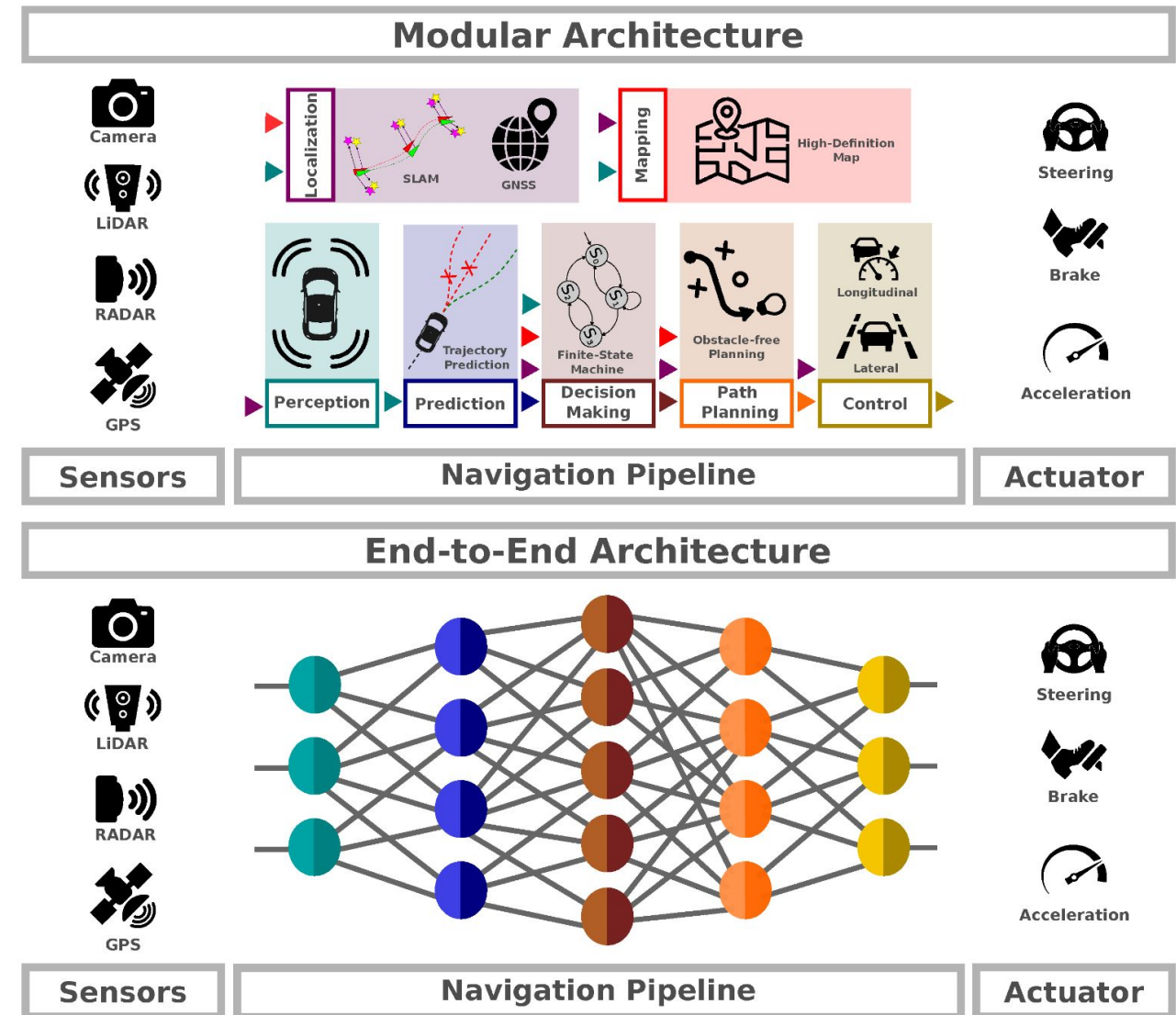


Figure 4. Modular & End-to-End Learning architecture illustration.[2]

Autonomous Driving System Paradigms

4. Hybrid Architecture

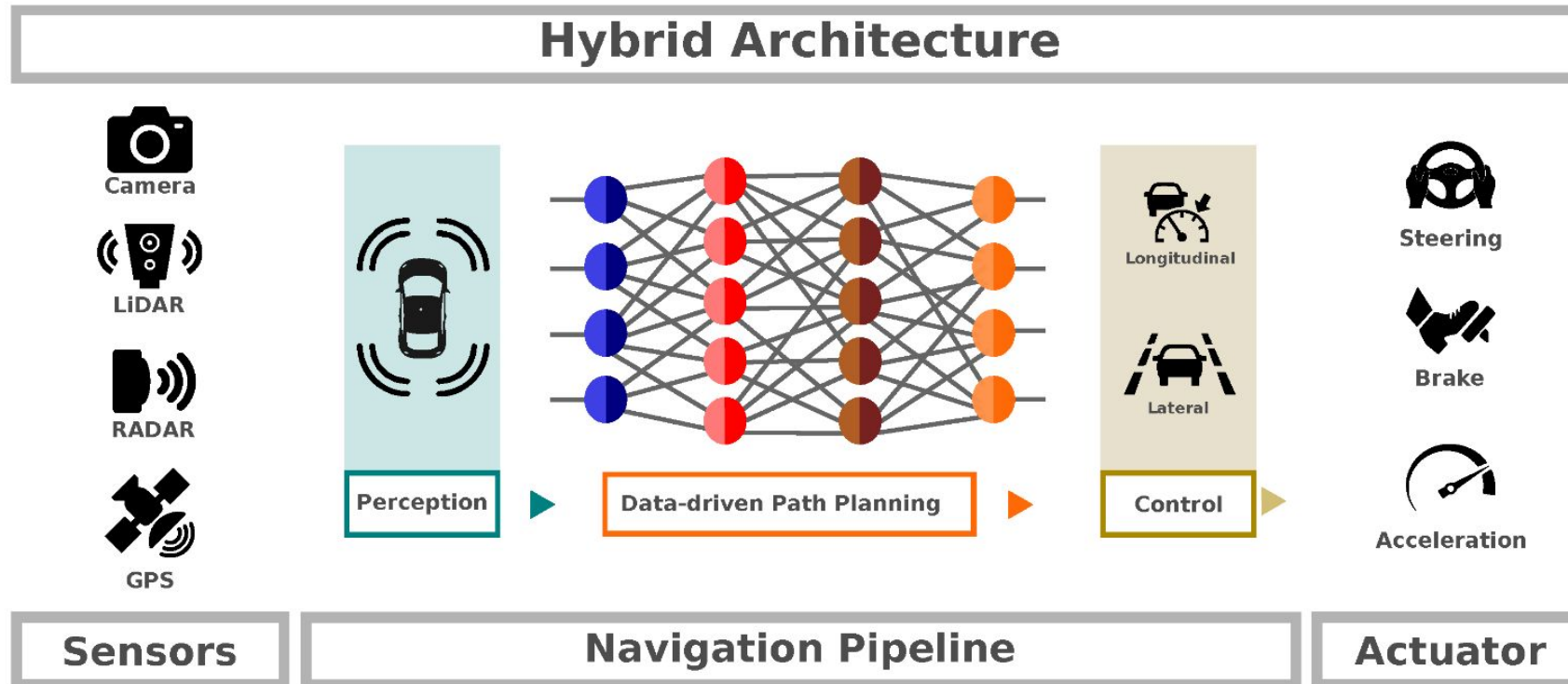


Figure 5. Hybrid architecture illustration.[2]

- Strengthened reliability due to modular handling of sensing and control.
- Used data-driven path planning for mapless environments to enhance flexibility and adaptability.
- It performs balance on map-based and dynamic, unknown scenarios.

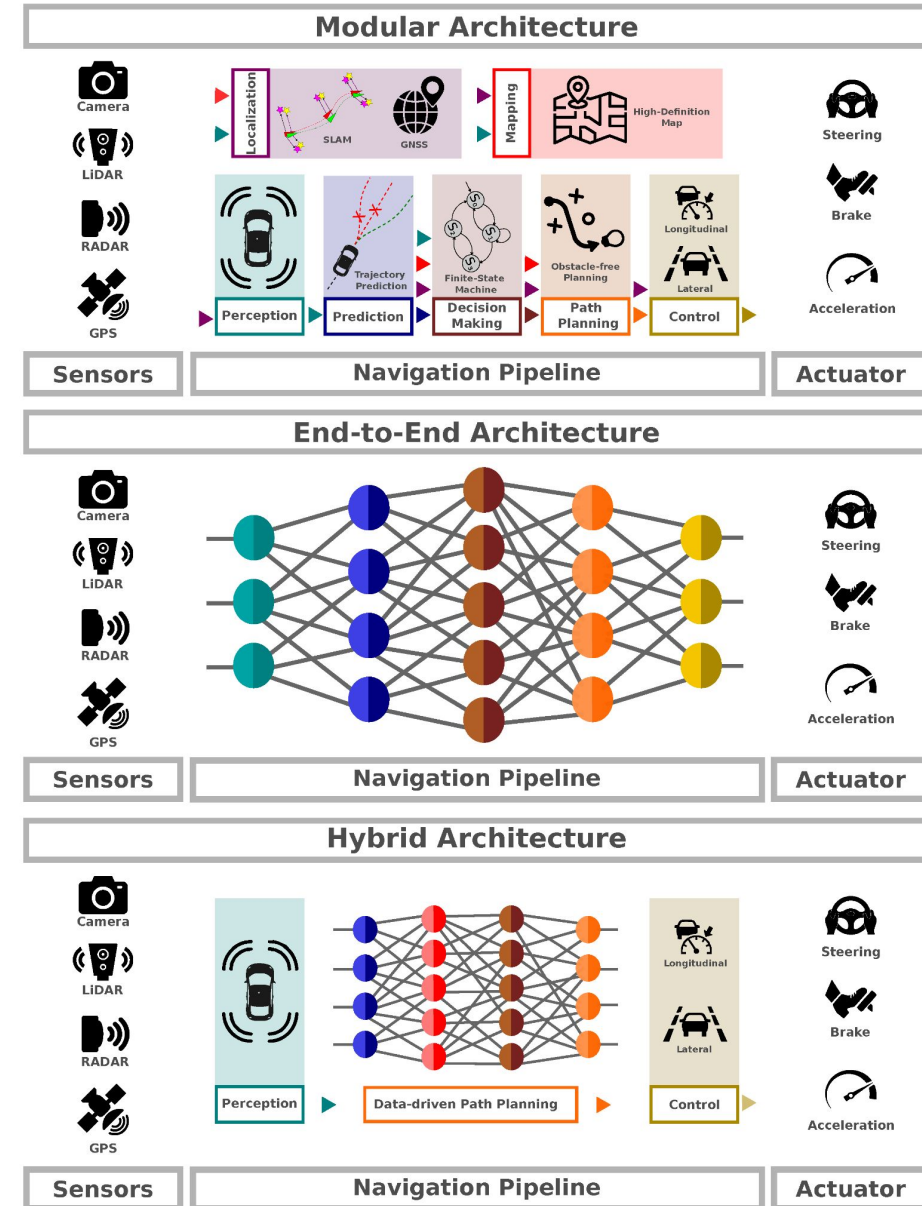
- *Explainable AI (XAI): Enhances trust and transparency in End-to-End Learning.*
- *Focus on creating robust, adaptable, and safe AD systems.*



- *Hybrid Architectures: Balancing interpretability and scalability.*

Summary

- *Modular systems excel in transparency and debugging.*
- *End-to-End systems are simpler and more adaptable but require extensive data.*
- *Hybrid approaches provide a balanced, scalable solution for diverse driving conditions.*



References:



- [1] Singh, S. (2018, March). Critical reasons for crashes investigated in the National Motor Vehicle Crash Causation Survey. (Traffic Safety Facts Crash•Stats. Report No. DOT HS 812 506). Washington, DC: National Highway Traffic Safety Administration.
- [2] L. A. G. I. P. d. S. J. A. R. P. C. A. W. D. F. & O. F. S. Rosero, “Integrating Modular Pipelines with End-to End Learning: A Hybrid Approach for Robust and Reliable Autonomous Driving Systems (<https://doi.org/10.3390/s24072097>),” Vols. Sensors, 24(7), 2097, 2024.
- [3] D. A. Pomerleau., “ALVINN: An Autonomous Land Vehicle In a Neural Network,” Advances in neural information processing systems 1. Morgan Kaufmann Publishers Inc., San Francisco, CA, USA,, p. 305–313., 1989.
- [4] Chart & Diagrams Design by [PresentationGO.com](https://www.presentationgo.com)